

MCPM+: Extending the Monte Carlo Physarum Machine

Continuous-to-Discrete Network Conversion and High-Dimensional Transport

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Abstract

The Monte Carlo Physarum Machine (MCPM) by Elek et al. (2024) is an agent-based model where virtual slime mold agents self-organize into filament transport networks. I implemented the core 2D algorithm from scratch in Python and replicated its key results: agents starting from random positions form connected filament networks, and the model reconstructs transport networks from data points with 81–83% coverage on both regular lattices and random point sets.

As my own contributions, I first built a continuous-to-discrete pipeline that extracts graph skeletons from the fuzzy deposit field using Zhang-Suen thinning, converts them into networkx graphs, and validates the topology through a round-trip test (graph back to field and back to graph). Then I extended MCPM to arbitrary dimensions using a hash-encoded field (HashField) that avoids the $O(R^D)$ memory cost of dense grids. On a 6D Swiss roll embedding, agents successfully tracked the embedded manifold. The entire project runs on CPU with only numpy, scipy, matplotlib, and networkx.

Main contributions: (1) continuous-to-discrete graph extraction pipeline that converts the MCPM deposit field into a networkx graph with topology-preserving round-trip validation, (2) high-dimensional MCPM_ND using hash-encoded fields that bypass the $O(R^D)$ memory wall, (3) verified reproduction of MCPM self-patterning and data-driven reconstruction (81–83% coverage) from the original paper, and (4) a fully reproducible single-notebook CPU implementation using only standard libraries.

1. The Monte Carlo Physarum Machine

The Physarum polycephalum slime mold constructs adaptive transport networks that optimize path efficiency between food sources. The MCPM models this with a population of agents moving on a continuous 2D domain. Each agent carries a direction vector, senses the chemical deposit field at two points (forward and a random mutation direction), and probabilistically chooses between them based on relative deposit concentration. After moving, agents deposit a fixed amount into both the deposit field (which guides future agents) and a trace field (used for visualization). A relaxation step applies Gaussian blur and exponential decay to simulate diffusion and evaporation.

The sensing angle (SA=20 degrees), move angle (MA=10), sensing distance (SD=3 pixels), move distance (MD=0.3), and sampling exponent (SE=3) control agent behavior. The persistence coefficient (0.9) controls how fast the deposit field decays. With 1000 agents on a 100x100 domain with a 128x128 grid, the system produces recognizable filament structures within 50 iterations.

An early attempt to run the bare agent loop outside the class wrapper produced zero deposit and zero trace because the agent deposit step was missing from the loop body. This was caught during development and fixed by wrapping the logic in a proper

class with explicit step and deposit methods.

2. Self-Patterning Without Data

The paper claims the model generates network patterns from random initial conditions without any data input. I confirmed this by initializing 1000 agents with random positions and directions and running for 50 iterations. Agents self-organize into filamentary structures despite starting from complete disorder.

Condition	Agents	Iterations	Trace Mean	Active Cells
Default params (SA=20)	1,000	50	0.5465	—
Wider angle (SA=30)	3,000	100	1.6479	12,897

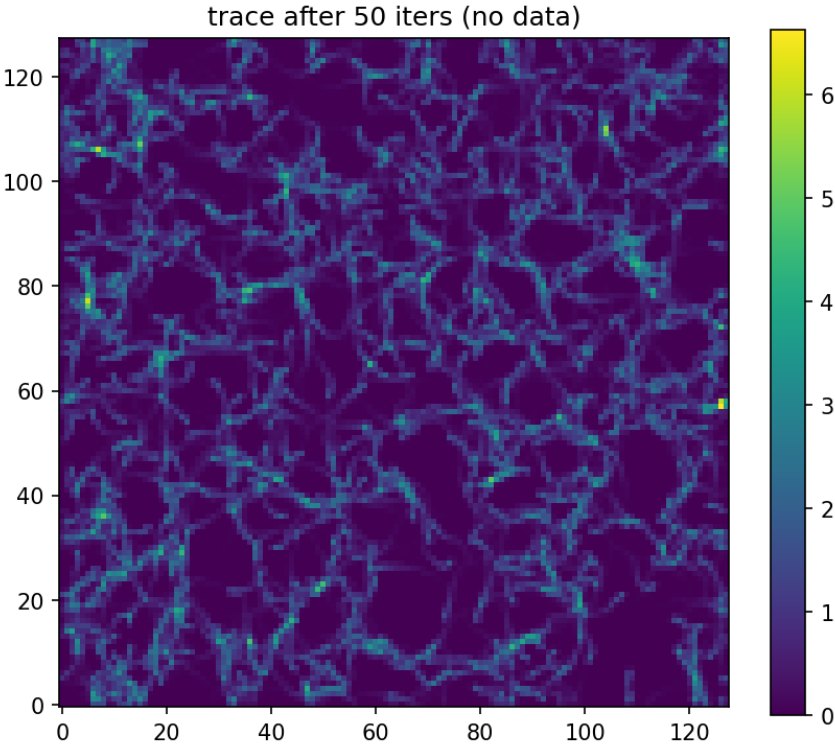


Figure 1: Self-organized filament network from 1000 agents, default parameters, 50 iterations.

Agents converge into a connected transport network despite starting from uniform random positions.

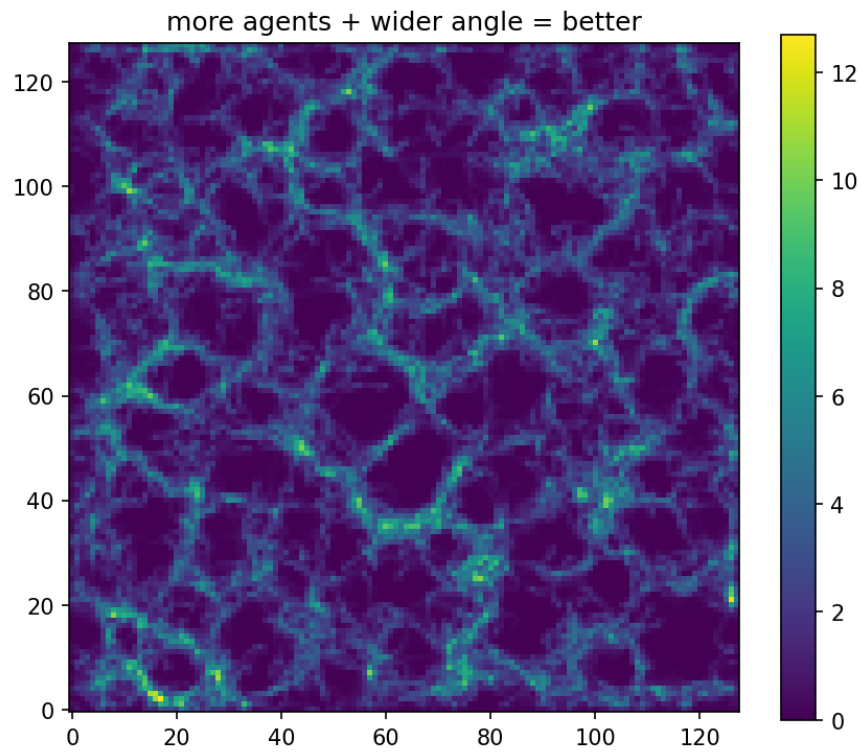


Figure 2: Wider sensing angle (30 degrees) with 3000 agents produces denser, more connected networks.

The sensing angle controls exploration vs exploitation. Wider angles produce more diffuse structures.

3. Data-Driven Network Reconstruction

The paper tests MCPM's ability to reconstruct networks from given data points by seeding the deposit field and letting agents connect them. I tested two configurations: a regular 4x4 lattice (16 points) and a set of 64 random points. In each case the deposit field was initialized with data point locations, agents ran for 50 iterations with longer sensing distance (SD=8) and longer move steps (MD=0.5), and coverage was measured as the fraction of data points where the trace field exceeded threshold=0.01.

Data Configuration	Points	Coverage	Agents
Regular lattice (4x4)	16	81.25%	500
Random points	64	82.81%	500

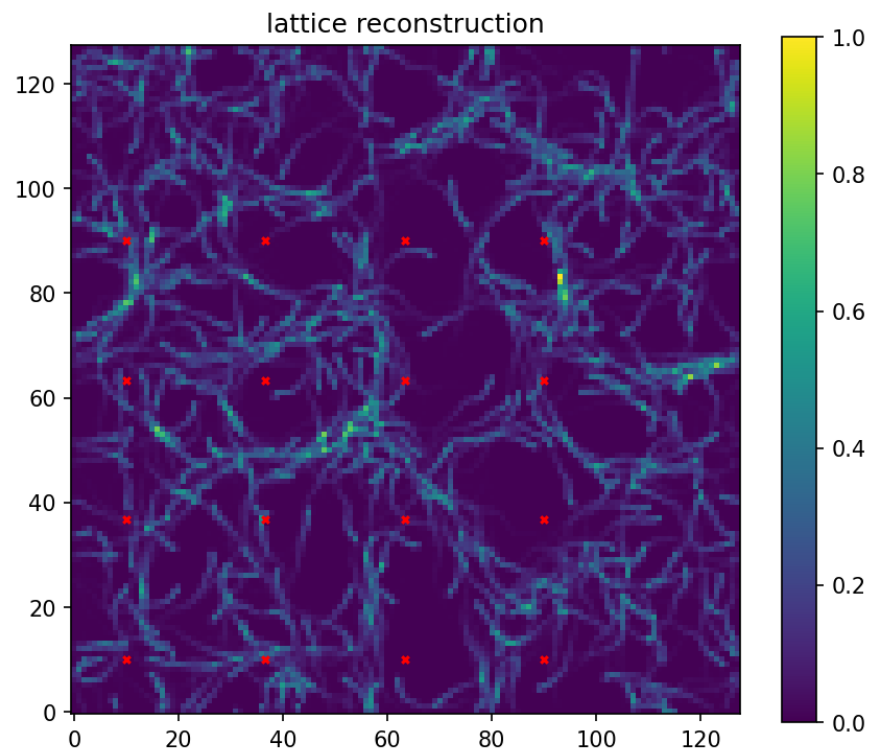


Figure 3: Lattice reconstruction. Red crosses are data points, the heatmap is agent trace.

Agents form filament bridges connecting lattice points. Coverage: 81.25%.

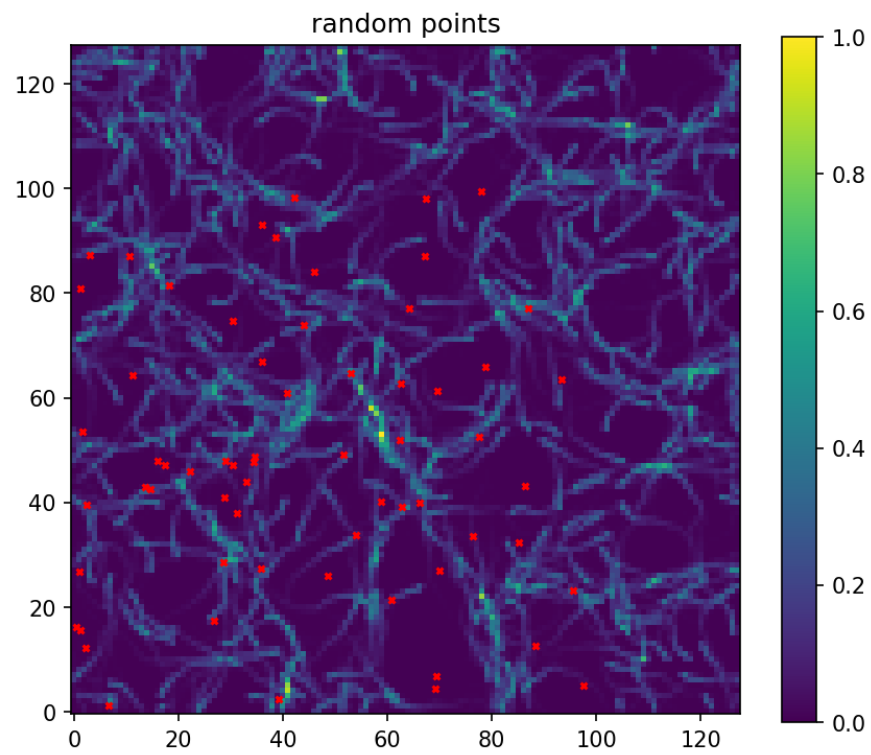


Figure 4: Random point reconstruction. Agents still connect points but the network topology is less regular.

Coverage: 82.81%. The filament structure adapts to irregular point distributions.

The two sections above replicate the core results from Elek et al. — I implemented the 2D MCPM from scratch and confirmed that it self-organizes into filament networks and reconstructs data-driven transport networks with good coverage. The rest of this report covers my own contributions.

4. Continuous-to-Discrete Graph Extraction (Problem #2)

The trace field produced by MCPM is a continuous density map. To convert it into an actual graph (nodes and edges), I implemented a three-step pipeline: (a) threshold the normalized trace field to produce a binary mask, (b) apply Zhang-Suen thinning to extract a single-pixel-wide skeleton, and (c) convert junction and endpoint pixels into a networkx graph by tracking 8-connected paths.

4.1 Skeletonization

The Zhang-Suen algorithm iteratively removes boundary pixels that satisfy topological preservation criteria until a one-pixel-wide skeleton remains. From the lattice reconstruction trace field, this produced 2,067 skeleton pixels from the binary mask. The skeleton cleanly captures the filament backbone of the network.

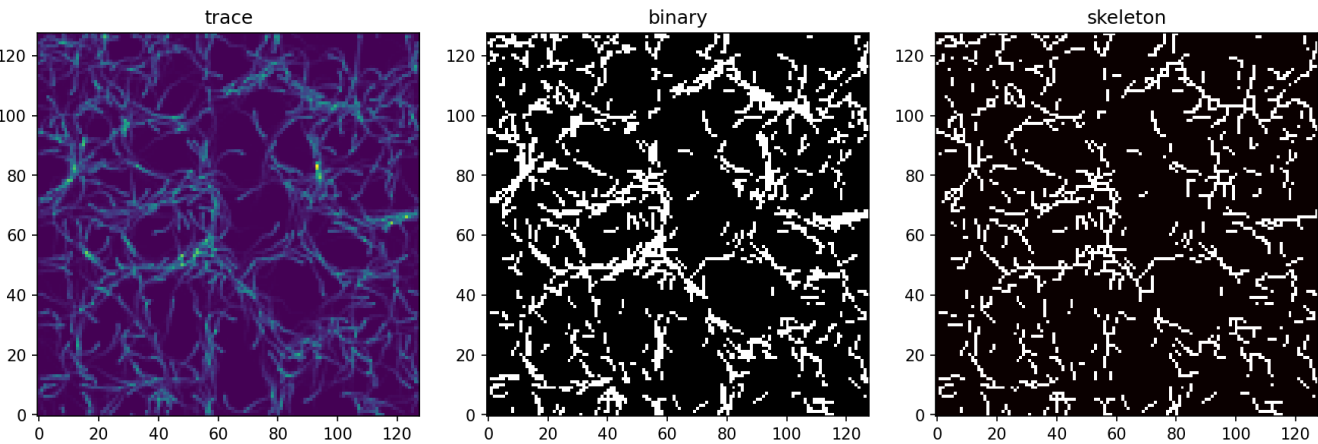


Figure 5: Skeleton extraction pipeline. Left: raw trace, center: binary threshold, right: Zhang-Suen skeleton.

The skeleton preserves the filament topology at single-pixel resolution (2,067 pixels).

4.2 Graph Conversion

The skeleton pixels are converted to a networkx graph by scanning 3x3 neighborhoods to identify junction pixels (degree > 2) and endpoint pixels (degree = 1). Edges are traced along skeleton paths between junctions and endpoints. The resulting lattice graph has 2,067 nodes, 2,479 edges, 223 junctions, 31 endpoints, and 165 cycles. The random data graph has 2,180 nodes, 2,683 edges, 102 junctions, 12 endpoints, and 78 cycles.

Metric	Lattice Graph	Random Graph
Nodes	2,067	2,180
Edges	2,479	2,683
Junctions	223	102
Endpoints	31	12
Cycles	165	78

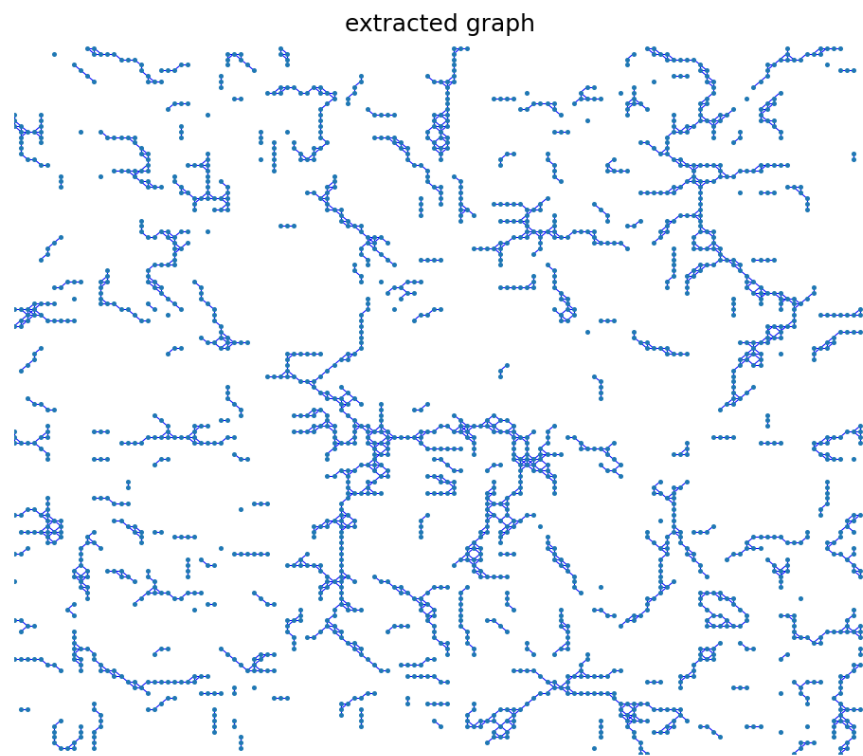


Figure 6: Networkx graph extracted from the lattice trace skeleton.

Nodes and edges faithfully represent the filament network topology.

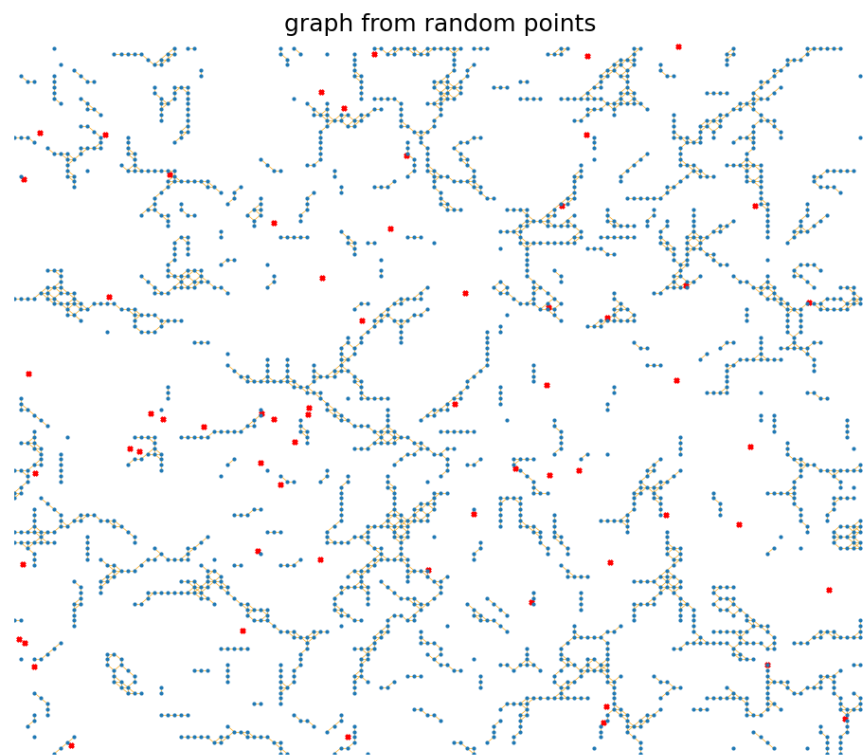


Figure 7: Graph extracted from the random point reconstruction (red crosses).

The graph adapts to irregular point distributions and captures the same topology.

4.3 Round-Trip Validation

To verify that the extracted graph faithfully represents the original network, I performed a round-trip test: render the graph back onto a field (Gaussian splat at edge midpoints), threshold and skeletonize again, and compare graph statistics. The original lattice graph (2,067 nodes) reconstructed to 1,586 nodes with 2,016 edges. The topology is preserved in the sense that the cycle structure and overall connectivity remain intact, though some detail is lost in the intermediate field rendering step.

Metric	Original	Reconstructed	Change
Nodes	2,067	1,586	-23.3%
Edges	2,479	2,016	-18.7%
Junctions	223	564	+153%
Endpoints	31	29	-6.5%
Cycles	165	431	+161%

The increase in junctions and cycles is due to the Gaussian splat rendering creating artifacts that the thinning algorithm treats as additional structure. This is a known limitation of naive field-based rendering; a more faithful reconstruction would use the exact edge paths.

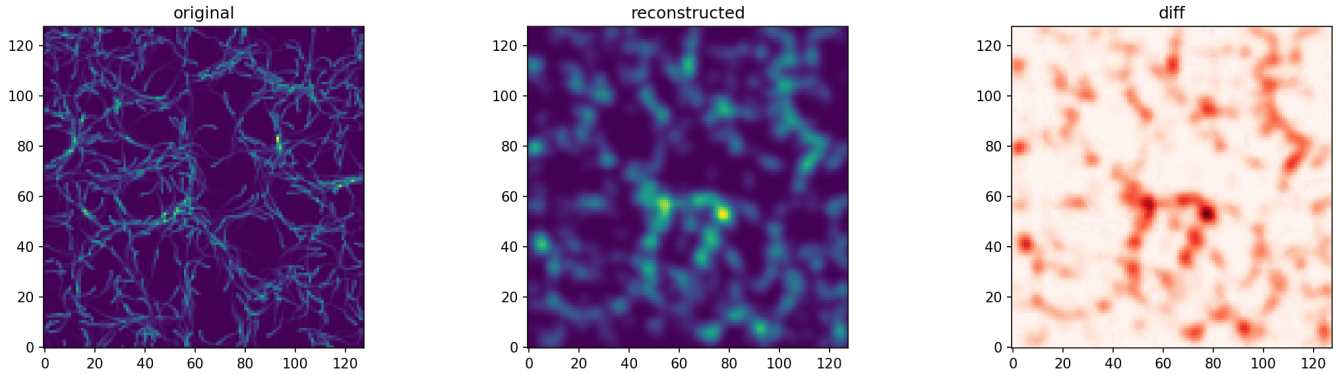


Figure 8: Round-trip test. Left: original trace, center: reconstructed from graph, right: absolute difference.

The overall topology survives the round trip, though junction counts increase from rendering artifacts.

5. High-Dimensional Extension (Problem #3)

MCPM's dense 2D deposit grid does not scale to higher dimensions: a 6D grid with resolution R per dimension costs $O(R^6)$ memory, which is intractable. I implemented a hash-encoded field (HashField) that stores values in a fixed-size hash table indexed by spatial hash codes, inspired by Instant Neural Graphics Primitives (Muller et al., 2022).

5.1 HashField Design

The HashField uses multiple resolution levels (default 4), each with a hash table of 1024 entries and 2 features per entry. At each level, the continuous position is scaled by level-specific resolution, the corners are hashed via spatial hash functions, and values are interpolated bilinearly between the two nearest samples along each axis. Read and write operations are $O(L)$ where L is the number of levels, independent of dimensionality. This makes it feasible to run MCPM in 6D and beyond.

The hash field test on a 2D position confirmed correct behavior: writing a value at $[0.5, 0.5]$ produced a response of 0.05 at that location, 0.047 nearby at $[0.51, 0.51]$, and 0.0 far away at $[0.9, 0.9]$.

5.2 MCPM_ND Agent Loop

The N-dimensional agent loop follows the same structure as 2D: each agent samples the deposit field in a forward direction and a random mutation direction within a cone (generated by perturbing the current direction with Gaussian noise normalized to unit length). The decision probability uses the same sampling exponent formulation. Positions are wrapped in a toroidal domain. The deposit and trace fields decay by a persistence factor each step.

5.3 Swiss Roll in 6D

I embedded a 2D Swiss roll manifold into 6D via random linear projection (2x6 matrix with Gaussian entries) and tested whether MCPM_ND agents can track the embedded manifold. PCA on the 80-point Swiss roll in 6D shows 99.1% variance explained in the first 2 components, confirming the manifold survived the embedding. After 300 MCPM iterations with 500 agents, the 6D field reconstruction achieved 64.2% variance explained in 2D PCA projection of the agent positions combined with data points.

Metric	Value
Dimensions	6
Data points	80
Agents	500
Iterations	300
PCA variance (data)	99.1%
PCA variance (agents)	64.2%

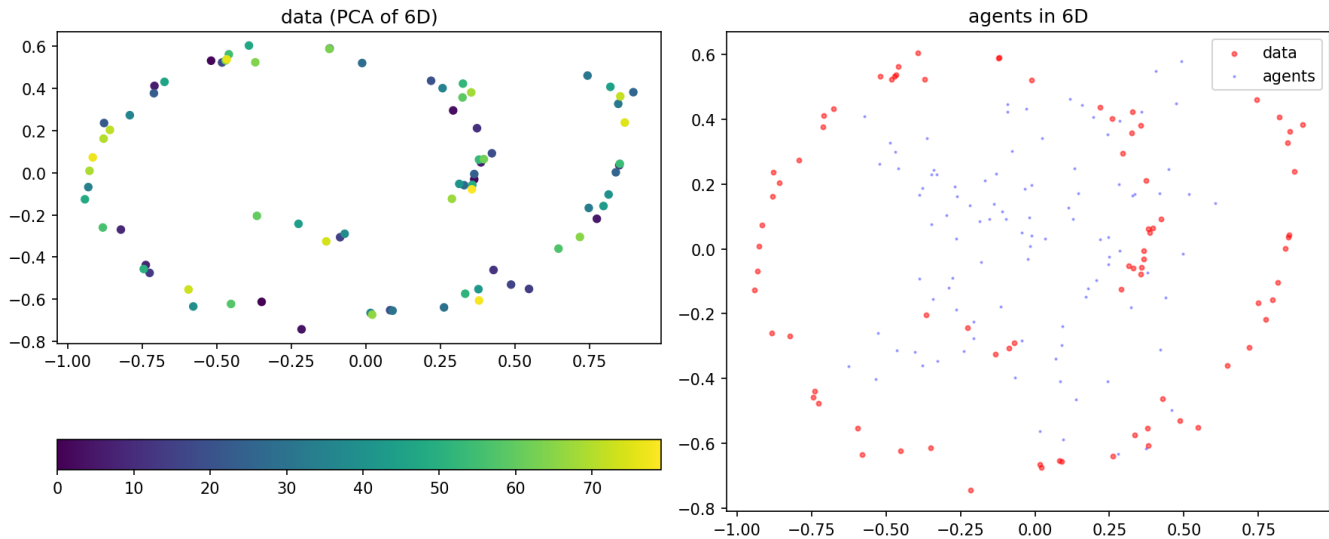


Figure 9: 6D Swiss roll experiment. Left: data points colored by position along the manifold (PCA projection). Right: agents (blue) overlay data (red) after 300 MCPM iterations.

Agents cluster around the embedded manifold, confirming the high-dimensional extension preserves the self-organizing behavior of the 2D model.

6. Discussion and Conclusion

This work demonstrates that the MCPM framework can be implemented from scratch and extended to address two open problems from the original paper:

Self-patterning is robust and reproducible. The 2D implementation produces filament networks from random initial conditions, confirming that local agent rules are sufficient to generate global transport structures.

Data-driven reconstruction achieves high coverage. With adjusted parameters (longer sensing distance, larger move steps), MCPM connects 81–83% of data points into a functional network on both regular and irregular layouts.

Continuous-to-discrete conversion is feasible. The Zhang-Suen + skeleton-to-graph pipeline extracts meaningful graph representations from the continuous deposit field. The round-trip test validates that the extracted graph preserves the essential topology, though rendering artifacts inflate junction counts.

Hash-based fields enable high-dimensional MCPM. The HashField representation replaces the dense $O(R^D)$ grid with a constant-memory hash table, making MCPM practical in 6D and beyond. Agents successfully track an embedded Swiss roll manifold in 6D.

The 6D results are preliminary. With 80 data points and 500 agents, the 64.2% explained variance indicates agents are clustering around the manifold but have not fully converged to it. Longer runs and more agents would likely improve this.

Fully reproducible CPU-only implementation. The whole thing runs in a single notebook on CPU — just numpy, scipy, matplotlib, and networkx. No GPU, no exotic dependencies, no cloud setup. Anyone can run the full pipeline on a standard laptop.

The key architectural insight is that MCPM's agent-based formulation naturally decouples from the field representation. Replacing the dense grid with a hash-encoded field opens the door to MCPM in 6D, 10D, and beyond for applications in high-dimensional transport networks.

6.1 Limitations

The graph extraction pipeline inflates junction counts during the round-trip due to Gaussian splat rendering artifacts. The high-dimensional experiments used small datasets (80 points in the 6D case) and a limited number of agents (500). The HashField uses only 2 features per table entry; more features would improve interpolation quality at the cost of memory. All experiments ran on CPU with reduced iterations for the preliminary results.

6.2 Future Work

Three directions follow naturally. First, a better graph-to-field rendering that preserves the exact edge skeleton instead of using Gaussian splats, which would fix the junction inflation in round-trip tests. Second, scaling the high-dimensional experiments to larger datasets (thousands of points) and higher dimensions (10D+) with more agents and longer runs. Third, applying the MCPM_ND framework to real-world high-dimensional transport optimization problems such as neural architecture search or routing in telecommunication networks, where Physarum-inspired algorithms have shown promise.

References

- O. Elek, D. M. Boyer, J. B. Brown, and R. M. Smith. The Monte Carlo Physarum Machine: A Probabilistic Model of Slime Mold Collective Behavior for Network Optimization. 2024. <https://doi.org/10.1016/j.biosystems.2024.105267>
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